Image Quality and Robustness in Face Recognition

A Comparison of Classical and Deep Approaches

Project Goal

Motivation

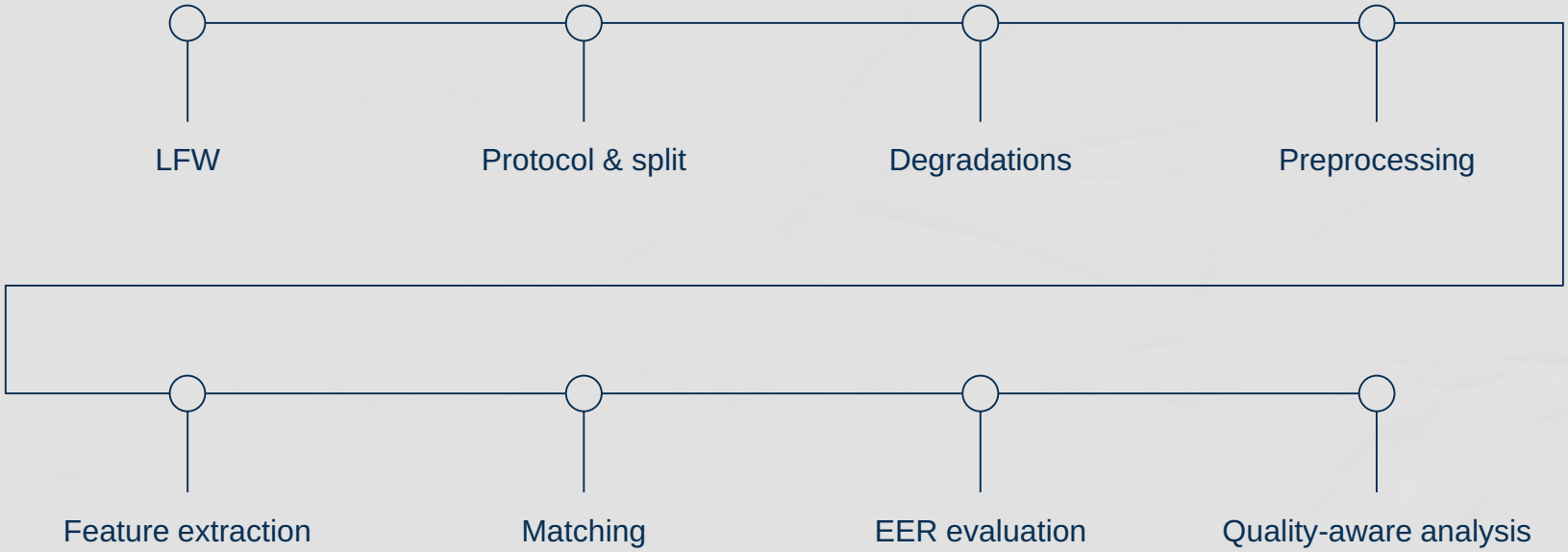
Face recognition performance depends not only on the recognition model, but also on the quality of the acquired biometric sample

Objectives

- Compare **Eigenfaces**, **LBPH** and **FaceNet**
- Evaluate robustness to controlled degradations
- Investigate **quality-aware rejection**

How does image quality affect different face-recognition approaches?

Pipeline



External: scikit-learn, OpenCV / NumPy, facenet-pytorch

Developed: experimental protocol, degradations, matching / evaluation, robustness and quality-aware analysis

Dataset & Experimental Protocol

Experiments are performed on the funneled version of the Labeled Faces in the Wild (LFW) dataset.

158 subjects – 10 selected images per subject – 1580 images used

Training

5 images / subject
790 images

Gallery

1 images / subject
158 images

Probe

4 images / subject
632 images

The three subsets are sample-disjoint: no image is shared between training, gallery, and probe sets.

The same identities appear in all three partitions, so the protocol is not subject-disjoint.

The gallery is kept clean, while degradations are applied only to probe images.

Verification Protocol and Evaluation Metrics

Every probe template is compared with every gallery template

Same identity ⟶ **genuine comparison**

Different identities ⟶ **impostor comparison**

632 genuine comparisons
99,224 impostor comparisons

All recognizers produce distance scores,
where a lower distance indicates a stronger match.

$$FAR(t) = \frac{FA(t)}{N_{impostor}}$$

Fraction of impostor comparisons incorrectly accepted

$$FRR(t) = \frac{FR(t)}{N_{genuine}}$$

Fraction of genuine comparisons incorrectly rejected

$$FAR(t_{ERR}) \approx FRR(t_{ERR})$$

The main performance measure is Equal Error Rate (EER): lower EER means better separation between genuine and impostor comparisons

Face Recognition Approaches

Eigenfaces

Global statistical representation

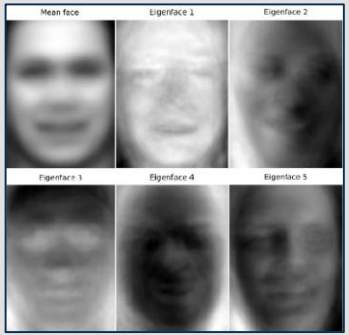
Training images

⇨ Mean face μ

⇨ Center x - μ

⇨ PCA

⇨ 171-D template



LBPH

Local hand-crafted representation

Local neighbourhood

⇩

Binary pattern

⇩

8x8 spatial grid

⇩

Local histograms

⇩

16,384-D template

FaceNet

Deep learned representation

160x160 RGB face

⇩

InceptionResnetV1

⇩

512-D embedding

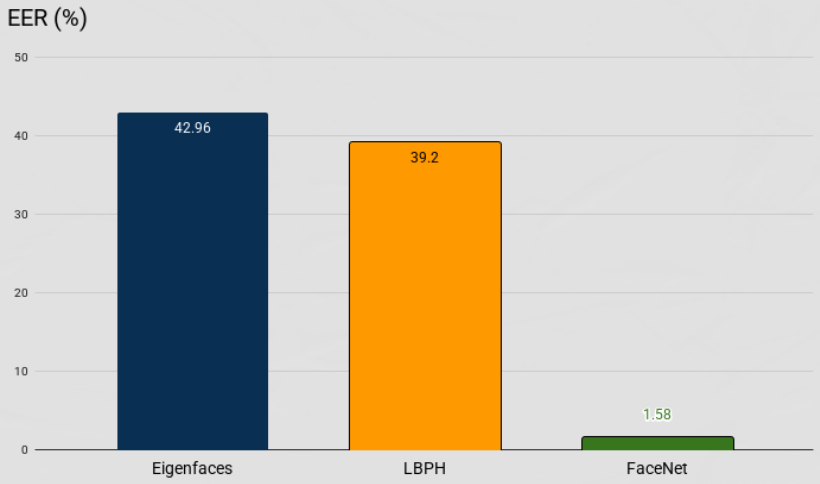
⇩

L_2 normalization

⇩

Euclidean matching

Clean-Image Baseline



Key observations

Eigenfaces and **LBPH** show limited discrimination under the adopted verification protocol, with EERs of **42.96%** and **39.20%**, respectively.

FaceNet performs substantially better, achieving an EER of only **1.58%**.

These clean-image results are used as the **reference baseline** for the **robustness experiments.**

Robustness Evaluation Protocol

The gallery remains clean, while each degradation is progressively applied only to the probe images

Clean gallery + *Degraded probe* ⟹ *Recognition* ⟹ *EER*

Gaussian Blur

$\sigma \in \{0, 0.5, 1.0, 2.0, 4.0\}$

Removes spatial detail and high-frequency information

Brightness

$\alpha \in \{0.5, 0.75, 1.00, 1.25, 1.50\}$

Modifies the global intensity distribution

Gaussian Noise

$\sigma \in \{0, 0.02, 0.05, 0.10, 0.20\}$

Introduces random pixel-level intensity perturbations

Resolution

$s \in \{1.00, 0.75, 0.50, 0.25, 0.125\}$

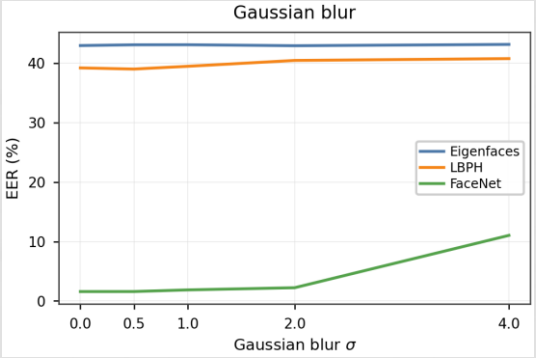
Removes spatial information through downsampling

Robustness to Blur and Noise

Gaussian Blur

FaceNet remains close to its clean performance under mild and moderate blur. At $\sigma = 4$, its EER increases from 1.58% to 11.03%.

LBPH and *Eigenfaces* vary only slightly, but both start from much weaker clean baselines.

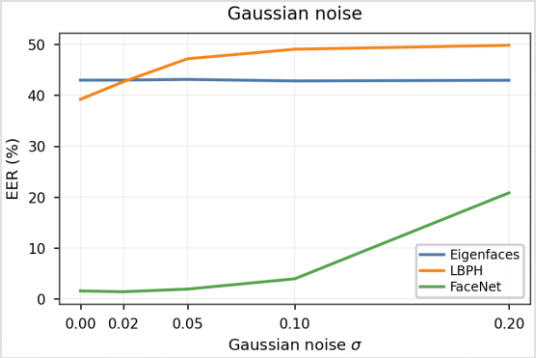


Gaussian Noise

Noise has the strongest effect on **LBPH**, whose EER rises from 39.20% to 49.80%.

FaceNet remains accurate under moderate noise, but reaches 20.83% EER at $\sigma = 0.20$.

Random pixel-level perturbations directly affect the local intensity patterns used by *LBPH.*



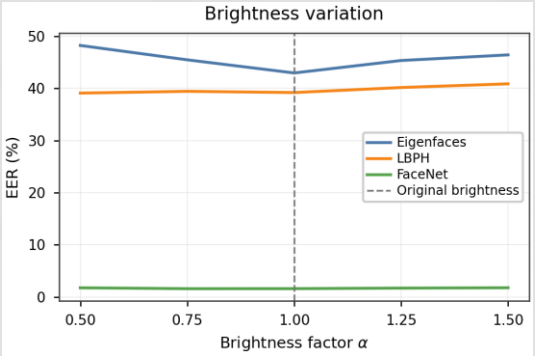
Robustness to Brightness and Resolution

Brightness Variation

FaceNet is almost unaffected across the tested brightness range, with EER values between 1.57% and 1.75%.

LBPH is also comparatively stable under global intensity changes.

Eigenfaces is more sensitive to brightness variation, reaching 48.23% EER at $\alpha = 0.50$.

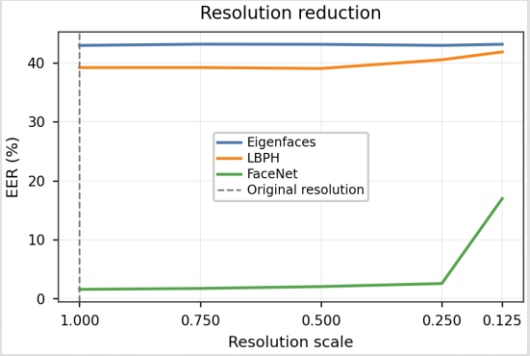


Resolution Reduction

FaceNet remains highly stable down to *s = 0.25*, where its EER is still only 2.57%.

At the strongest reduction, *s = 0.125*, its EER rises sharply to 16.98%.

Extreme downsampling removes spatial detail required for reliable matching.



Overall Robustness Comparison

	Eigenfaces	LBPH	FaceNet
Clean	42.96%	39.20%	**1.58%**
Blur $\sigma = 4$	43.15%	40.76%	**11.03%**
Noise $\sigma = 0.20$	42.94%	49.80%	**20.83%**
Brightness $\alpha = 0.50$	48.23%	39.09%	**1.74%**
Resolution $s = 0.125$	43.15%	41.85%	**16.98%**

Key observations

LBPH is particularly vulnerable to Gaussian noise, remaining comparatively stable under brightness variation.

Eigenfaces is most clearly affected by global brightness changes.

FaceNet achieves the lowest absolute EER in every tested condition, although severe blur, noise and resolution loss still produce substantial degradation.

A small change in EER does not necessarily mean better robustness if the clean baseline is already poor.

$$\Delta EER = EER_{degraded} - EER_{clean}$$ ⇨ Absolute **EER** and **ΔEER** should therefore be interpreted together

Quality-Aware Recognition

Quality-Aware Strategy

Low-quality probes are detected and rejected before matching instead of forcing a recognition decision for every acquisition.

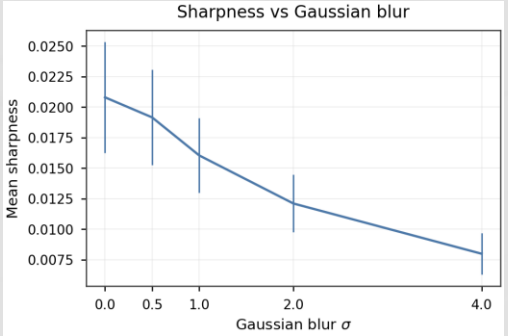
Probe

⇨ Sharpness score

⇨ Reject low-quality probes

⇨ Recognition

Sharpness as Quality Measure



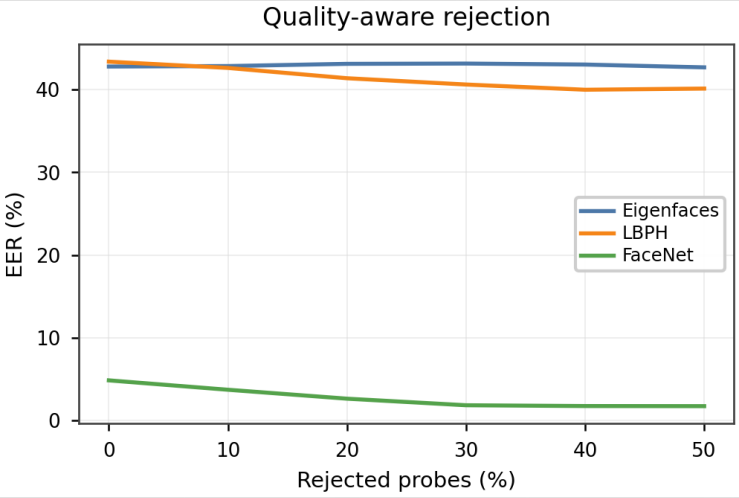
Increasing blur clearly reduces sharpness

Clean: 0.0208 ⟶ $\sigma = 4$: 0.0080

Mixed-quality set: 5 blur levels x 632 probes = 3,160 probes

Rejection is based on measured sharpness, not on the known blur level

Quality-Aware Rejection Results



FaceNet

4.85%* → *1.84%

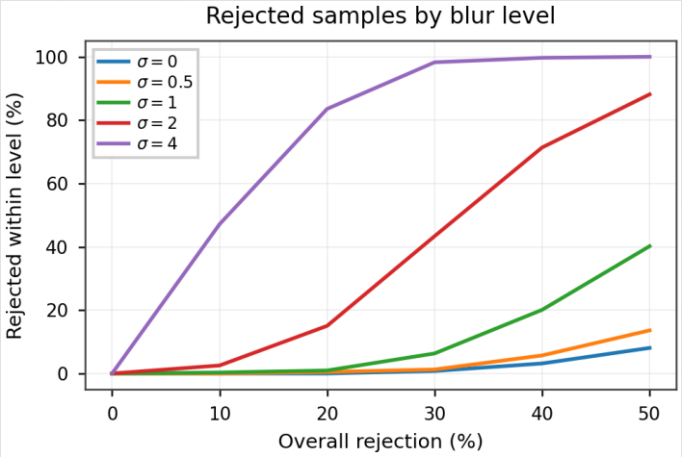
at 30% rejection

Clean-image baseline: **1.58%**

Rejecting the lowest-quality 30% of probes removes most of the degradation responsible for the increased FaceNet error.

Beyond 30%, further rejection provides only marginal improvement.

What is being rejected?



At higher rejection rates, progressively more acceptable samples are also discarded

At 30% overall rejection:

Severe blur - $\sigma = 4$
98.26% rejected
621 / 632 probes

Moderate blur - $\sigma = 2$
43.35% rejected
274 / 632 probes

Clean probes - $\sigma = 0$
0.79% rejected
5 / 632 probes

The rejection mechanism is therefore strongly concentrated on *severely blurred samples* rather than removing probes indiscriminately.

Limitation: sharpness is effective for blur, but it is not a general Face Image Quality Assessment measure

Conclusions

1) Different representations react differently to image-quality degradations

Classical, hand-crafted, and deep representations exhibit different robustness behaviours under changing acquisition conditions.

2) Strong recognition does not imply complete robustness

FaceNet remains the strongest approach overall, but severe degradation can still substantially reduce recognition reliability.

3) Quality information can be used before matching

A quality-aware strategy can identify unreliable acquisitions and improve the reliability of the recognition process.

Thank you!

Questions?

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